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 [Final Project](#) / UCSD ExtraSensory Dataset 📱

UCSD ExtraSensory Dataset 📱

This dataset was collected right here on the UCSD campus. It contains in-the-wild smartphone and smartwatch telemetry from 60 users, originally built to see how well mobile devices can recognize everyday behavioral context (you can read the original paper [here](#)).

The good news is that the heavy lifting of raw signal processing is already done. Instead of dealing with raw audio waves or accelerometer arrays, you get a tabular dataset of pre-computed features— such as battery drain rates, location variance, app usage, and audio MFCCs—paired with context labels reported by the users themselves.

Getting the Data

You'll need to grab the primary features and labels archive directly from the [ExtraSensory Project Page](#).

`ExtraSensory.per_uuid_features_labels.zip` (215 MB) [Download Link](#)

Reading the Data: Unlike the other project options, this isn't a single file. The archive contains 60 separate `.csv.gz` files (one for each user, named by their anonymized ID). You'll want to write a quick loop using `os` or `glob` to read a subset of these files and concatenate them into a single pandas DataFrame.

You can read the zipped files directly into pandas without extracting them first: `pd.read_csv(filepath, compression='gzip')`

Column	Description
<code>uuid</code>	Anonymized ID of the user (you can extract this from the filename)
<code>timestamp</code>	Unix time of the recorded 1-minute window
<code>label:SLEEPING</code>	Indicator (0, 1, or NaN) of whether the user was sleeping
<code>label:IN_CLASS</code>	Indicator (0, 1, or NaN) of whether the user was in class
<code>raw_acc:magnitude_stats:mean</code>	Mean magnitude of the accelerometer over the window
<code>discrete:app_state:is_active</code>	Indicator of whether an app was actively being used
<code>audio_naive:mfcc0:mean</code>	Pre-computed audio features (Mel-frequency cepstral coefficients). <i>Note: You don't need to process raw audio; these are just standard numerical columns!</i>
<code>lf_measurements:loc_variance</code>	Spatial variance of the user's location

(Note: There are over 200 feature columns and 51 label columns. You'll definitely want to use `df.filter(regex=...)` or column slicing to subset what you actually care about early in your notebook).

If you want to pull in extra data like mood or absolute GPS coordinates, check out the supplementary zip files on their site.

Idea Generation: Questions and Prediction Problems

The questions below are **broad categories and thought-starters**. We expect you to formulate your own specific questions, hypotheses, and prediction tasks based on what you find during your Exploratory Data Analysis. Do not just copy these verbatim; use them to inspire your own unique angle!

Steps 1-4: Exploration and Testing

- **Label Missingness:** Users constantly forget to manually tag what they are doing, leaving you with `NaN` values in the label columns. Think about whether the missingness of a specific label depends on passive sensor states (e.g., is the user's phone dead, or are they driving?).
- **Behavioral and Contextual Shifts:** Do passive sensor readings (like "spatial entropy" or audio variance) change significantly when users report being in different states (like socializing versus studying)?
- **Digital Habits:** Are there statistical relationships between how a user interacts with their phone OS (switching apps, screen unlocks) and their physical activity levels later in the day?

Steps 5-8: Predictive Modeling

- **Forecasting Behavior:** Set up a regression model to predict a continuous metric (like a user's total active minutes or sleep duration for a given day) based on rolling averages of their telemetry from previous days.
- **A Focused Classifier:** While the dataset has 51 different context labels, trying to predict all of them at once will likely negatively impact your accuracy and make the project overly complicated. Instead, pick a focused subset. You could build a classifier to distinguish between 2 or 3 distinct states that interest you. Similarly pick features that are relevant, not everything.
- **Privacy-Safe Context Prediction:** Try predicting a specific user context (like commuting or studying) using *only* a restricted subset of "innocent" background sensors (e.g., battery drain and screen status), ignoring highly sensitive data like GPS and microphone features.

A Few Tips

- **Don't crash your kernel:** If you try to merge and process all 60 users at the 1-minute interval level right away, pandas might hang. Start by building your pipeline on 5 to 10 users. Use `.resample('H')` or `.groupby()` to roll the 1-minute data up into hourly or daily summaries before you do any heavy processing.
- **Pick your sensors:** You don't get bonus points for throwing all 225 columns into a model. Pick a specific family of sensors (like just the motion sensors, or just the phone state variables) that makes sense for the narrative you are building.
- **NaNs are data too:** In mobile telemetry, missing data usually tells a story. If a phone stopped recording location or a user stopped answering surveys, think about *why* before you just use `.dropna()`.